ECON6067 Individual Final Project

Cash Transfers and Local General Equilibrium

Replication of Egger et al. (2022) Table V, with an extension on payment timing and nearby treated villages

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Repository	github.com/StupidKevin7743/ZhenyuYang_3036056641

Paper approval pending. Egger et al. is outside the four illustrative options in the project brief. Instructor approval is still required for this paper and the proposed replication scope.

The repository includes the code, summary results, data-access instructions, and an AI-use disclosure. The research datasets and restricted coordinates must be obtained separately from an authorized source.

1. Main findings

Does giving cash to poor households also raise incomes elsewhere in the local economy? Egger et al. (2022) study a one-time program that gave about USD 1,000 to each of more than 10,500 households in rural Kenya. Treatment was randomized across a study area of 653 villages. The paper reports that each dollar transferred generated about USD 2.50 of additional local economic activity, with little price inflation. This project checks the multiplier estimates in Table V; it does not independently re-estimate the paper's price effects.

The replication gives an expenditure multiplier of **2.58** and an income multiplier of **2.47**, matching the paper after rounding. Their average is **2.52**. Across all 11 table quantities, the largest difference between the Python and saved Stata estimates is less than 0.0000002. A completed 2,000-draw Python bootstrap gives similar, though slightly larger, standard errors. The estimates support a positive local response, but the evidence that the multiplier exceeds one is weaker.

The extension checks two practical questions: did some types of village start receiving payments earlier, and are there enough villages without nearby treatment to make simple comparisons? Within subcounties, the timing regression detects no clear association with the three measured village characteristics. At 2 km, however, only **five of 653 villages** have no treated neighbor; at 3 km and 4 km, none do. Simple comparisons of four treatment/exposure groups would therefore rely on very small or empty groups.

Scope and evidence

- Core: Table V estimates, their components, and a comparison of bootstrap uncertainty.
- Extension: descriptive checks of payment timing and village exposure groups.
- Implementation: completed Python computation checked against Stata point estimates; the direct 2,000-draw Stata run remained incomplete.

These findings concern the selected table and extension. They do not certify every result in the article or establish a new causal effect from the extension.

2. Research question and economic mechanism

When a household receives cash, it may buy food from a local shop, pay a builder, or purchase farm inputs. Those sellers earn income and may spend some of it locally. People who never received a transfer can therefore benefit too. The research question is how much additional spending and income this process creates across the study area. These wider effects are what the paper calls local general-equilibrium effects.

The **transfer multiplier** is the extra local economic activity divided by the cash transferred. A value of 2.5 means that USD 100 transferred is associated with an estimated USD 250 increase in activity over the measured period. It does not mean that each recipient receives another USD 250 in cash. The estimate covers recipients, other households, and firms, using the paper's expenditure or income accounting.

Why Table V

Table V brings the results together in two ways: it adds extra household spending and business investment, and it separately adds extra wages, profits, capital income, and taxes. These are two measures of the same broad response, so they should not be added together. Their average is a third summary. The paper's Figure 1 shows the quarterly paths; this project reproduces Table V and uses Figure 1 only as context.

Why the size of the effect is uncertain

Economic channel	What it means for the multiplier
Spending circulates locally	A shopkeeper or worker can spend new income again, spreading gains beyond recipients.
Firms can produce more	Unused capacity can allow higher sales and output without equally large price increases.
Purchases leave the area	Imports and spending elsewhere can reduce the local response.
The response takes time	Adding effects across 29 months captures more than a single survey-date comparison.

The extension asks whether the data provide useful comparisons for studying these effects. A village receiving no cash may still have treated neighbors, so it is not automatically an unaffected comparison village.

3. Data, sample, and experimental design

The study includes 653 villages in 84 sublocations, with 328 villages assigned to treatment and 325 to control. Randomization occurred in two stages. First, sublocations or groups of sublocations were assigned to 33 high-saturation and 35 low-saturation groups. Then two-thirds of villages in the high groups and one-third in the low groups were assigned to treatment. This created random variation in both a village's own treatment and treatment nearby.

Households living under thatched roofs were eligible. In treated villages, eligible households received KES 87,000: KES 7,000 initially, KES 40,000 two months later, and KES 40,000 eight months after the first payment. The census covers 65,383 households. Baseline and endline surveys cover 7,845 and 8,239 households; these are study-wide survey counts, not the sample size of every Table V regression. The median endline interview was 19 months after experimental start. Enterprise and market surveys supply business and price measures.

Inputs used by this package

Logical input	Role	Public-repository treatment
Household/enterprise multiplier file	Spending and income components for Table V	Access instructions only; never committed
Stata multiplier estimate file	Point-estimate cross-check	Generated/authorized local input; never committed
Experimental timing panel	Actual rollout month	Access instructions only; never committed
Household/village analysis files	Endline-weighted size proxy, eligibility, treatment	Access instructions only; never committed
Provider-supplied village-buffer file	Counts of nearby treated villages	No coordinates committed; input remains untracked

Data access and scope

The repository distributes summary tables, figures, and this report; it does not distribute research observations or coordinates. The input manifest lists the seven required files. The project starts from prepared analysis files, because the earlier raw-data construction code contains record-specific material and is excluded. Restricted-GPS analyses elsewhere in the paper remain outside this project's scope.

4. Identification and replication method

A simple comparison of recipients and nonrecipients would miss gains reaching nearby households and firms. Equation (7) in the paper therefore relates each outcome to transfers paid in the village and within 2 km of it, in the current quarter and each of the previous nine quarters. The explanatory variables are quarter-specific transfer amounts, not cumulative transfers to date.

The instrumental-variables (IV) design uses randomized assignment and rollout to isolate transfer variation that is not chosen in response to local outcomes. The paper then adds the estimated spending and income responses over 29 months. Durable assets and inventories are measured as stocks at endline. Prices and transfers are converted to January 2015 values, so the calculation is in real terms.

Core workflow

Stage	Implementation	Validation
1. Estimate	Compute each spending and income component in Python	Compare 11 quantities with saved Stata estimates
2. Measure uncertainty	Run 2,000 wild-cluster bootstrap draws	Use seed 12345 and common signs within sublocations
3. Compare	Calculate totals, averages, and bootstrap tail frequencies	Show differences from published Table V
4. Present	Generate summary tables and figures directly from code	Check numbers and inspect rendered pages

Inference

The wild-cluster bootstrap repeats the calculation after multiplying regression residuals by random +1 or -1 signs, using the same sign for observations in a sublocation. Each artificial outcome combines the fitted value with the signed residual. The spread across 2,000 draws gives the bootstrap standard error. Following the supplied code, the reported one-sided quantities are the shares of draws below zero or one. They are not probabilities that a hypothesis is true.

Numerical comparison rule

The point-estimate tolerance is 0.000001. Matching those estimates checks the coefficient calculations; it does not prove that the whole bootstrap is identical to Stata. Standard errors and tail frequencies are therefore compared separately with the paper, and the differences remain visible in the results.

Why not use the earlier three-draw result?

Three draws can produce only very coarse tail frequencies and unstable standard errors. The earlier run shows that the code executed, but it is unsuitable for final inference. This report uses the completed 2,000-draw Python results.

5. Core replication results

All 11 point estimates match the paper to two decimal places. The expenditure multiplier is 2.5767, income is 2.4691, and their average is 2.5229. Their bootstrap standard errors are 1.4575, 1.7874, and 1.4278, respectively: slightly larger than the paper's 1.44, 1.71, and 1.39. Thus, the central estimates are very close, while the uncertainty estimates are similar rather than identical.

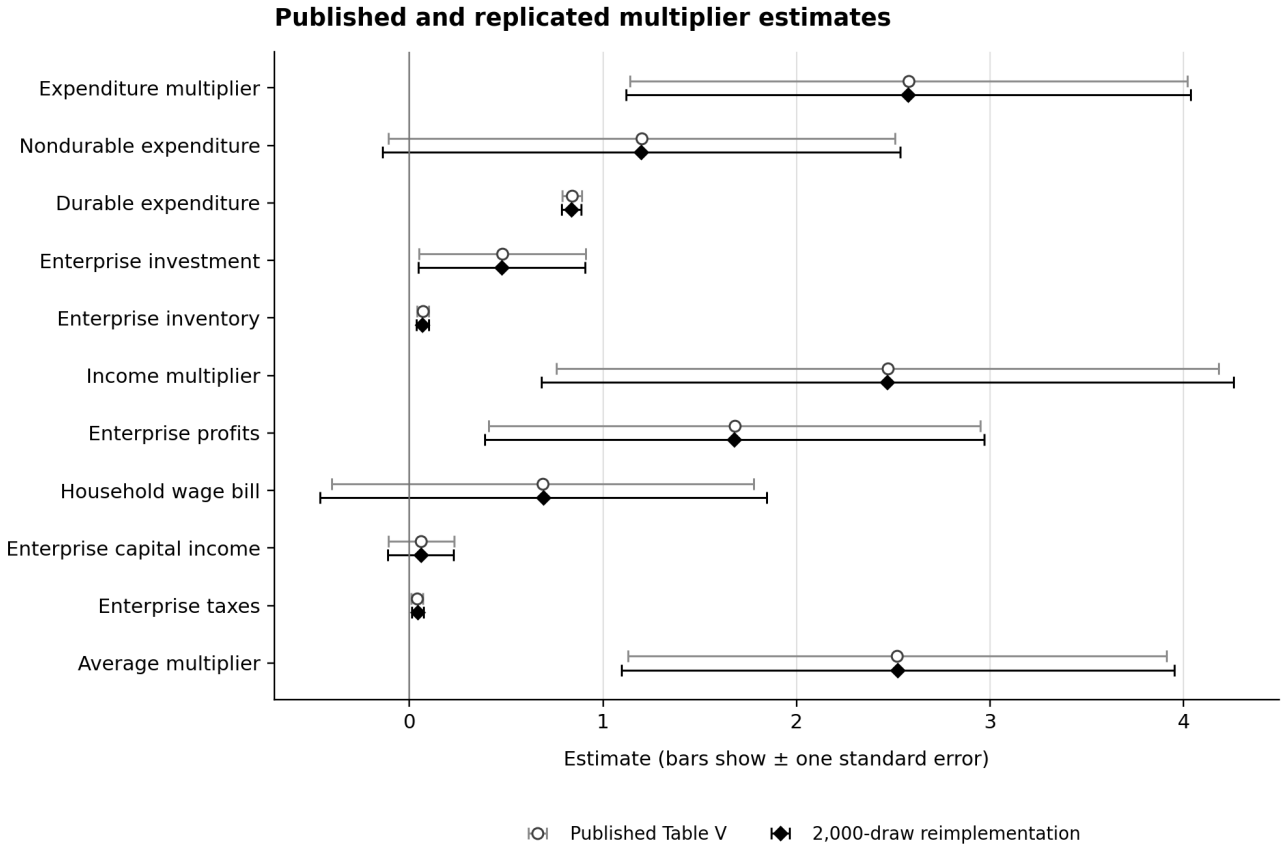
Component	Paper	Python	Paper SE	Python SE	Below 0	Below 1
Expenditure multiplier	2.58	2.58	1.44	1.46	0.0305	0.1455
Household non-durable expenditure	1.20	1.20	1.31	1.34	0.1905	-
Household durable expenditure	0.84	0.84	0.05	0.05	0.0000	-
Enterprise investment	0.48	0.48	0.43	0.43	0.1435	-
Enterprise inventory	0.07	0.07	0.03	0.03	0.0115	-
Income multiplier	2.47	2.47	1.71	1.79	0.0860	0.2105
Enterprise profits	1.68	1.68	1.27	1.29	0.1035	-
Household wage bill	0.69	0.69	1.09	1.15	0.2740	-
Enterprise capital income	0.06	0.06	0.17	0.17	0.3630	-
Enterprise taxes paid	0.04	0.04	0.03	0.03	0.0955	-
Average of both multipliers	2.52	2.52	1.39	1.43	0.0360	0.1495

Table 1. Published Table V and Python results. SE = bootstrap standard error. The last two columns are the fractions of bootstrap draws below 0 and 1, using the supplied code's convention. A dash indicates an unreported component-level test. A displayed 0.0000 means no draw fell below the threshold, not zero true probability.

Interpretation

The point estimates are economically large, but they are imprecise. For expenditure, 3.05% of bootstrap draws fall below zero and 14.55% below one; for income, the corresponding shares are 8.60% and 21.05%. The shares in which **both** multipliers fall below the threshold are 1.30% for zero and 7.85% for one. These are the code's joint-tail measures. Under the paper's reporting convention, the evidence for a positive response is stronger than the evidence for a multiplier above one.

6. How closely do the results match?



Report Figure 1. Published and Python Table V estimates. Each horizontal line extends one standard error either side of its point estimate; these are not 95% confidence intervals. Marker shapes distinguish the series in black-and-white print.

Point-estimate cross-check

Across 11 quantities, the largest absolute difference between Python and saved Stata point estimates is 0.00000017844. This is below the 0.000001 tolerance. The two implementations therefore give effectively the same central estimates at the precision reported in the paper.

Why some uncertainty estimates differ

The Python and published standard errors differ slightly. Different random-number streams and numerical routines may contribute, but this project has not isolated the exact source of each difference. One supplied-code behavior is retained: the untreated-household asset coefficient is held fixed during resampling. This is documented for reproducibility, not endorsed as the only valid choice. A completed original-code Stata bootstrap would provide a stronger comparison.

7. Extension: timing and comparison groups

Did some types of village receive payments earlier? The first check relates the start month to village size, the share of eligible households, and high-saturation assignment. It controls for average timing differences between subcounties, so the comparison is within each subcounty. Standard errors allow errors to be correlated within sublocations. Village size and eligibility shares use endline household weights: this is a descriptive timing check, not a baseline balance test.

Start is measured in two ways: the first month with an initial KES 7,000 payment, and the first month when cumulative initial payments reach 10% of the village's total initial payments. Coefficients are in months; positive values indicate later starts.

Timing model	Covariate	Coef.	SE	p-value
First token	Log of weighted village size	-0.064	0.205	0.755
First token	Eligible-household share	-0.207	0.995	0.835
First token	High-saturation assignment	0.168	0.286	0.557
10% cutoff	Log of weighted village size	-0.047	0.202	0.818
10% cutoff	Eligible-household share	-0.084	1.017	0.934
10% cutoff	High-saturation assignment	0.185	0.287	0.519

Table 2. Timing diagnostics: 328 treated villages and 80 sublocation clusters. Subcounty indicators are included. SEs are cluster-adjusted; p-values use a two-sided normal approximation. The eligible share is measured from 0 to 1.

Timing result

None of the three characteristics has a statistically detectable association with start month in either specification. For example, high-saturation assignment is associated with a 0.168-month later first payment (SE 0.286; p = 0.557). This is consistent with little association in these data, but a nonsignificant result does not establish that actual payment timing is random.

Which comparison groups exist?

The second check separates own-village treatment from the presence of at least one other treated village nearby. The labels describe assignment and proximity; they do not prove that an economic spillover occurred.

Own village	Treated neighbor?	2 km	3 km	4 km
Treated	No	1	0	0
Treated	Yes	327	328	328
Control	No	4	0	0
Control	Yes	321	325	325

Table 3. Village counts. Each radius includes the same 653 villages; the focal village is excluded from its own neighbor count.

8. Why nearby treatment matters

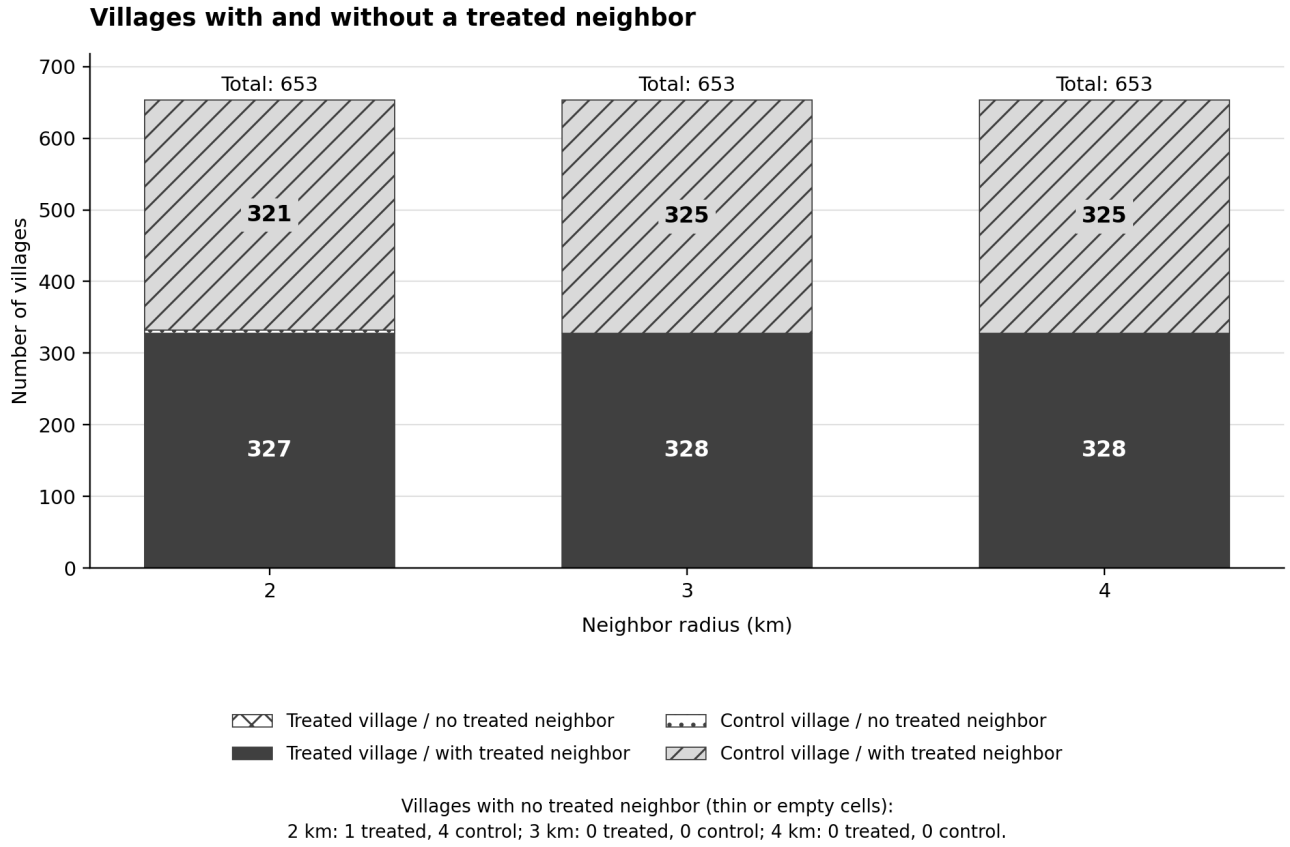


Figure 2. Treatment and nearby-treatment groups. Each bar contains all 653 villages. The very small groups retain their true scale; exact counts are in Table 3 and below the chart.

Almost every village has a treated neighbor

At 2 km, villages have an average of 13.88 neighbors. Only one treated village and four control villages have no treated neighbor. These two groups account for just 0.3% and 0.9% of weighted households. At 3 km and 4 km, every village has a treated neighbor.

An average based on one or four villages would be sensitive to those particular villages. Empty groups cannot be compared at all. This explains the difficulty of a simple four-group analysis and the value of examining **how much** cash was transferred nearby. The counts do not, by themselves, validate the paper's instruments or show that spillovers caused an outcome change.

Limits of the extension

Payment timing is monthly, the regressions cover only three characteristics, and the exposure groups reduce nearby transfers to yes/no categories. These checks describe the data and comparison groups; they do not estimate a new causal effect.

9. Reproducibility package

The repository keeps the original Stata/R analysis code and adds a Python workflow for this project's results. Install the dependencies in requirements.txt and place the seven authorized input files at the paths listed in config/input_manifest.json. Then run **python3 code/run_all.py --target all** from the repository root to regenerate the tables and figures. Input paths can also be supplied explicitly.

Package map

Path	Purpose
code/run_all.py	Run the core and extension; check expected inputs and outputs
code/table5_multiplier_bootstrap.py	Table V point estimates and 2,000-draw Python bootstrap
code/extension_timing_exposure.py	Timing regression, village-group counts, and exposure chart
code/render_figures.py	Table V comparison chart from summary CSVs
code/build_report.py	Build report.pdf from the summary tables and figures; requires ReportLab
config/	Project status, input contract, and published benchmarks
data/raw/ and data/processed/	README access instructions only; inputs remain untracked
outputs/tables/ and outputs/figures/	Summary results used by this report
skills/SKILL.md	Reusable instructions for running and checking the project
AI_USE_DISCLOSURE.md	Stage-by-stage AI assistance and human verification

How to check the package

Run **python3 code/run_all.py --check-only** to check the public files without estimating the models. The full computation fixes the bootstrap at 2,000 draws and seed 12345. The release validator checks that only the approved summary outputs are included. It does not replace numerical or visual review. The PDF is built separately using **python3 code/build_report.py**.

The full project workflow completed using the existing local analysis files. Reproducing the numbers on another machine still requires authorized access to those inputs. A fresh-machine run of the full original article has not been completed.

10. Validation, limitations, and AI use

Validation completed

- The source paper and appendix were checked against their contents, including Table V and the experimental design.
- The core computation used 2,000 draws and seed 12345. Its summary has 11 estimate rows and one joint-tail row.
- All 11 Python point estimates agree with the saved Stata estimates within 0.000001.
- The timing and village-group results agree with the earlier local analysis. The final editorial revision leaves the numerical CSVs unchanged.
- Tables, charts, and the rendered PDF were checked for numerical consistency, readable labels, and page layout.

Substantive limitations

- The direct 2,000-draw Stata loop remained incomplete. Point-estimate agreement does not establish exact equivalence of every bootstrap step.
- The supplied code holds one untreated-household asset coefficient fixed during resampling. This behavior is retained and should be tested in further work.
- Household surveys begin about nine months after rollout and enterprise surveys about 18 months after rollout. Early quarterly responses are therefore less precisely estimated.
- The multiplier depends on the paper's accounting and horizon. Imports, asset depreciation, and activity outside the study area limit its interpretation as a full welfare or national-GDP effect.
- The village-size and eligibility covariates use endline weights, and the exposure groups simplify continuous transfer amounts. Neither extension check establishes a new causal effect.
- Restricted-GPS analyses and other article exhibits are outside the project scope. Alternative-paper approval remains pending.

AI use and human responsibility

OpenAI Codex assisted with reading the instructions, organizing the repository, implementing and checking Python code, drafting the report, and revising its text and figures. Source material and machine-generated results provide the evidence. The student remains responsible for reviewing the methods, explaining the results, and deciding what to submit. AI_USE_DISCLOSURE.md gives the full account.

11. Conclusion

The replication gives an expenditure multiplier of 2.58 and an income multiplier of 2.47, matching the published figures after rounding. They suggest that one dollar transferred generated about USD 2.50 of additional local economic activity over the measured period. The estimates remain uncertain: the results support a positive response more clearly than they establish a multiplier above one.

The extension shows why a simple treated-versus-control comparison can be misleading in this setting. Almost every village has a treated neighbor. At 2 km, only one treated village and four control villages have no treated neighbor; at 3 km and 4 km, none do. Comparisons built around these tiny groups would be unreliable. The counts help explain the paper's use of nearby transfer amounts, while leaving the causal effect of those transfers to the original IV design.

The contribution is a transparent check of Table V and a practical diagnosis of the comparison groups. The remaining substantive work is to compare a completed Stata bootstrap and test the retained resampling choice. Instructor approval is also required before using this alternative paper for the final submission.

References

Egger, D., Haushofer, J., Miguel, E., Niehaus, P., and Walker, M. (2022). General Equilibrium Effects of Cash Transfers: Experimental Evidence From Kenya. *Econometrica*, 90(6), 2603-2643. https://doi.org/10.3982/ECTA17945.

Egger et al. Complete Appendices A-J (7 June 2022): Appendix A for the design and Appendix D for multiplier sensitivity. Main-paper references: Section 2 for randomization, Section 3.8 and Equation (7) for dynamics, and Table V and Section 5 for the multiplier and its limitations.

Rubric map

Project criterion	Report location
Understanding and motivation	Sections 1-2
Data sources, sample, variables	Section 3
Core replication and validation	Sections 4-6
Independent extension	Sections 7-8
Reproducibility/workflow	Section 9 and repository
Limitations, AI disclosure, conclusion	Sections 10-11 and AI_USE_DISCLOSURE.md